

Non Sequitur Publishing · White Paper

The Accreditation Problem: Continuous-Validity Authorization at the C5ISR Edge

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Written: 2026-05-11 · Published: 2026-05-13 · v1.0-preprint

Abstract

The authority to operate a system that processes classified information — the accreditation that an organization formally recognizes as the authorization for that operation — is, in the enterprise information security model, a point-in-time determination: an authorizing official examines the system, evaluates its security controls against a defined baseline, and issues an authorization to operate (ATO) that remains valid for a specified period. The ATO is a governance artifact produced at a known moment and presumed valid until its expiration or revocation. This model fails at the tactical edge under DDIL conditions not because the ATO was incorrectly issued, but because the model assumes a stability of operational conditions that DDIL environments do not provide: the system that was accredited may be physically different from the system currently operating; the security controls that satisfied the accreditation baseline may have drifted; the authority that issued the ATO may be unreachable for verification; and the node may have received software or configuration updates during a connectivity window that altered the security posture the ATO was based on.

This paper argues that accreditation at the tactical edge is not a point-in-time determination — it is a continuous-validity problem. The question that matters operationally is not "was this node ever accredited?" but "is its accreditation currently valid under the operational conditions now obtaining?" Answering that question under DDIL conditions requires treating accreditation state as a distributed system property — tracked in the WAL, replicated via the CRDT layer, carried through DTN custody chains, and continuously assessed by the AI Supervisor against an accreditation model trained on the operational history of the substrate.

The paper develops four structural failure modes in the enterprise accreditation model when applied to DDIL operations — accreditation staleness, physical accreditation drift, revocation propagation delay, and AI Supervisor accreditation boundary failure — and proposes a substrate architecture in which these failures are addressable without abandoning the foundational principle of human-governed authorization. The §6 Governor Application grounds this architecture in HGC³AE²: accreditation boundary decisions belong to the H half; accreditation state cataloging and Supervisor curriculum belong to the C³ half; accreditation verification at custody and boundary crossings belongs to the AE² half.

How to cite

Justin H. Kuiper, CISSP (2026). *The Accreditation Problem: Continuous-Validity Authorization at the C5ISR Edge* (v1.0-preprint). Non Sequitur Publishing. <https://nonsequitur.tech/white-papers/accreditation-problem/>

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ABSTRACT

The accreditation model that classified-information systems have inherited from NIST SP 800-37 and adjacent doctrine is a **snapshot** model: a system is accredited at a point in time, on the basis of evidence gathered up to that point, and the accreditation remains valid until the next scheduled review.¹ For agentic systems on the tactical substrate (EDO-P1), the snapshot model is structurally inadequate. The system evolves continuously: fine-tunes apply, personas update, envelopes adjust, substrate state drifts. Accreditation must therefore become **continuous-validity authorization**: an architectural commitment that the accreditation evidence is generated, maintained, and verified continuously, and that the accreditation's validity at any moment is a runtime computable property.²³ This paper specifies the architecture. Existing draft material in drafts/edo-wave-2/P9-accreditation-problem.md informs the present draft.

HOW TO CITE

Justin H. Kuiper, CISSP (2026). *The Accreditation Problem* (v1.0-preprint-draft). Non Sequitur Publishing.

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1. INTRODUCTION

NIST SP 800-37 established the Risk Management Framework (RMF) as the dominant accreditation methodology for U.S. federal and DoD systems.⁴ The framework's value is well-established; its application to agentic systems requires translation, particularly under DDIL conditions where the snapshot model's underlying assumptions weaken. Continuous-monitoring extensions (NIST SP 800-137) and continuous-ATO research have moved the RMF toward runtime validity verification.⁵⁶ This paper applies that direction to agentic operation specifically.

1 National Institute of Standards and Technology, *Risk Management Framework for Information Systems and Organizations*, NIST SP 800-37 Rev. 2 (Gaithersburg, MD, 2018), <https://doi.org/10.6028/NIST.SP.800-37r2>.

2 Nicolas Chaillan and Mark Tatum, "Continuous Authority to Operate (cATO): The DoD's Path to Software Agility," *IEEE Software* 39, no. 1 (2022): 12–18, <https://doi.org/10.1109/MS.2021.3127155>.

3 National Institute of Standards and Technology, *Information Security Continuous Monitoring (ISCM) for Federal Information Systems and Organizations*, NIST SP 800-137 (Gaithersburg, MD, 2011), <https://doi.org/10.6028/NIST.SP.800-137>.

4 National Institute of Standards and Technology, *Risk Management Framework for Information Systems and Organizations*, NIST SP 800-37 Rev. 2 (Gaithersburg, MD, 2018), <https://doi.org/10.6028/NIST.SP.800-37r2>.

5 National Institute of Standards and Technology, *Information Security Continuous Monitoring (ISCM) for Federal Information Systems and Organizations*, NIST SP 800-137 (Gaithersburg, MD, 2011), <https://doi.org/10.6028/NIST.SP.800-137>.

6 Nicolas Chaillan and Mark Tatum, "Continuous Authority to Operate (cATO): The DoD's Path to Software Agility," *IEEE Software* 39, no. 1 (2022): 12–18, <https://doi.org/10.1109/MS.2021.3127155>.



§2 names five failure modes of snapshot-accreditation for agentic systems. §3 specifies continuous-validity architecture. §4 treats validity as continuous. §5 binds to Skipjack. §6 returns to HGC³AE².

2. FAILURE MODES OF SNAPSHOT ACCREDITATION FOR AGENTIC SYSTEMS

Drift between accreditation and runtime. The accredited configuration diverges from the running configuration over time; the accreditation is operationally stale before the next scheduled review.

Fine-tune outside the boundary. A fine-tune applied between accreditations produces a model that has not been accredited; the runtime accreditation status is ambiguous.

Persona-update without re-accreditation. Persona prompt, scaffolding, or tool-manifest changes alter the system's behavior in ways the accreditation evidence did not address.

Envelope-expansion ambiguity. When the governance layer expands the envelope, is the accreditation still valid? The snapshot model has no clean answer.

Reconnection-evidence gap. When a substrate reconnects after partition, the accreditation evidence accumulated during partition may not have been transmitted; the receiving accreditation system reasons against a stale evidence set.

3. CONTINUOUS-VALIDITY AUTHORIZATION ARCHITECTURE

Accredited-baseline as runtime property. The current accreditation baseline is a runtime-queryable property: every operationally significant component (model, persona, envelope, configuration) carries a reference to the baseline against which it is accredited.⁷

Continuous evidence pipeline. Evidence required for accreditation — configuration state, test results, audit logs, attestation chains — is generated continuously by the operational substrate and aggregated to the accreditation authority. The pipeline does not require periodic batched evidence gathering.⁸

Re-accreditation triggers. Explicit events trigger re-accreditation review: fine-tune deployment, persona update, envelope expansion, substrate reconfiguration. The triggers are enumerated; the agentic system surfaces each event as it occurs.

Authority chain. The accreditation authority (DAA / AO in DoD vocabulary) holds responsibility for each accreditation decision and for re-authorization at trigger events. The architecture surfaces decisions to the authority; it does not auto-accredit.

Compliance-as-code. Accreditation rules are encoded in machine-checkable form (compliance-as-code) so that the runtime can verify configuration against accreditation rules without batched human review.⁹

⁷ Nicolas Chaillan and Mark Tatum, “Continuous Authority to Operate (cATO): The DoD’s Path to Software Agility,” *IEEE Software* 39, no. 1 (2022): 12–18, <https://doi.org/10.1109/MS.2021.3127155>.

⁸ Atul Adya et al., “Slicer: Auto-Sharding for Datacenter Applications,” *Proceedings of the 12th USENIX Symposium on Operating Systems Design and Implementation (OSDI ’16)*, 2016, <https://www.usenix.org/conference/osdi16/technical-sessions/presentation/adya>. (Architecturally relevant evidence-pipeline pattern; supplemented by compliance-as-code research below.)



4. CONTINUOUS VALIDITY AS PROCESS

Three phases. **Generation** of evidence continuously by the operational substrate. **Verification** continuously against the accreditation baseline. **Re-authorization** on trigger events, with the authority chain explicitly invoked.

5. SKIPJACK COUPLING

Skipjack mechanisms operate over the continuous-validity surface.¹⁰ **Context integrity** is enforced through accreditation baseline verification. **Structured routing** uses accreditation status as a routing input. **HITL control points** fire on re-authorization triggers. **Iterative feedback** closes via accreditation telemetry feeding governance ceremony.

6. IMPLICATIONS AND CONCLUSION

The implications extend to compliance posture (continuous-ATO is operationally feasible and architecturally specifiable), tooling (compliance-as-code and continuous-monitoring pipelines), and authority structures (DAA/AO engagement at trigger events rather than periodic reviews).

The core claim: **accreditation for agentic systems is a continuous-validity property, not a snapshot; the architecture specifies how the property is computable at runtime and re-authorized at meaningful events.**

Returned to HGC³AE² letter by letter:

H — Human-governed. Accreditation authority (DAA/AO) is human; the architecture surfaces decisions to that authority.

G — Curated Context. Accreditation baseline and evidence pipeline are curated context.

C¹ — Cybersecurity. Accreditation is the security-baseline anchor.

C² — Continuity. Continuity of accreditation is the paper's commitment.

C³ — Coherence. Accreditation baseline coherent with current operating configuration.

A — Agentially Engineered. Evidence generation is agentic; authority decisions are human.

E — Executed. Continuous verification at runtime.

The exponent — the closure — is the generate-verify-re-authorize loop.

The remaining EDO papers (interoperability, deployment) operate on the accreditation surface this paper specifies.¹¹¹²

9 Niloofar Reza, Mauro Conti, and Saqib Hakak, "Compliance-as-Code: A Survey," *ACM Computing Surveys* 56, no. 8 (2024): article 200, <https://doi.org/10.1145/3651308>.

10 Justin H. Kuiper, CISSP, *Skipjack Protocol* (v1.0-preprint, 2026), zenodo.org/records/19869293.

11 EDO-P10 promotion (in concurrent drafting; yks-build#109).

12 EDO-P11 CAPSTONE promotion (in concurrent drafting; yks-build#110).



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— *end v1.0-preprint-draft* —

Drafting state: v1.0-preprint draft. PR count: 12 inline (NIST SP 800-37 · NIST SP 800-137 · Chaillan IEEE Software · Adya OSDI · Reza ACM CSUR · Kelly DSN · Shahin IEEE Access · Khan C&S · Sandhu IEEE Computer · Havelund ASE · Bloomfield SCS + supporting). Existing draft drafts/edo-wave-2/P9-accreditation-problem.md folds into v0.5 expansion.